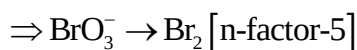
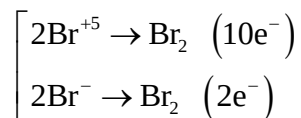
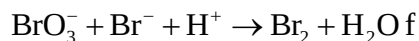


Solutions

(Redox-Reactions)

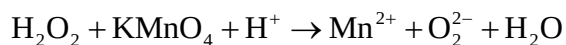
Question 1.



$$\text{Then equivalent weight} = \frac{\text{molecular mass}}{5}$$

Then ratio of molecular mass and equivalent mass = 5

Question 2.



$$(\text{NV})_{\text{KMnO}_4} = (\text{NV})_{\text{H}_2\text{O}_2}$$

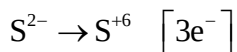
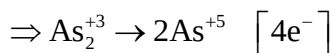
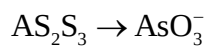
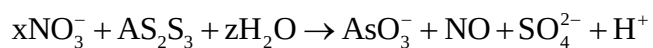
$$= \text{normality of KMnO}_4 \Rightarrow 5 \times 0.1 \Rightarrow 0.5$$

$$(0.5 \times 10) = 28 \times N$$

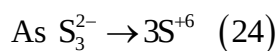
$$N = 0.1785$$

$$\text{Then volume \%} = 0.1785 \times 5.6 = 1$$

Question 3.



Then total n-factor

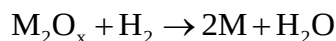
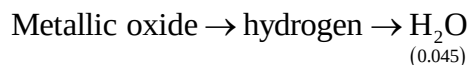


Then n-factor = 28

Then x = 28

Then $\frac{x}{2} = \frac{28}{2} = 14$

Question 4.

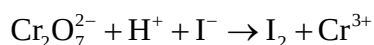
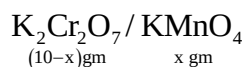


Then $\left(\frac{W}{E}\right)$ for metal oxide = $\left(\frac{W}{E}\right)\text{H}_2\text{O}$

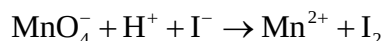
$$\Rightarrow \left(\frac{0.22}{E+8}\right) = \frac{0.045}{9}$$

$$\Rightarrow E = 36 \Rightarrow \frac{E}{9} = \frac{36}{9} = 4$$

Question 5.



Then $\frac{W}{E_{\text{Cr}_2\text{O}_7^{2-}}} \times 1000 = N \times V$



$$\frac{W}{E_{\text{KMnO}_4}} \times 1000 = N \times V$$

$$\Rightarrow \text{equation mass of } \text{K}_2\text{Cr}_2\text{O}_7 \Rightarrow 31.6$$

$$\Rightarrow \text{equation mass of } \text{KMnO}_4 \Rightarrow 49$$

$$\text{Then } \frac{(10-x)}{49} \times 1000 + \frac{x}{31.6} \times 1000 = 2.2 \times 100$$

$$10 \left[\frac{10-x}{49} + \frac{x}{31.6} \right] = 2.2$$

$$\frac{10-x}{49} + \frac{x}{31.6} = 0.22$$

$$\Rightarrow 31.6(10-x) + 49x = 0.22 \times 49 \times 31.6$$

$$316 - 31.6x + 49x = 340.64$$

$$X = 1.41$$

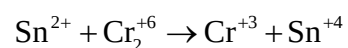
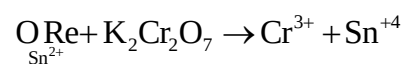
$$\text{Then mass \% of KMnO}_4 = \frac{1.41}{10} \times 100 = 14.16$$

$$\text{Then } \frac{27}{5} = \frac{2 \times 14.16}{5} = 5.66 \approx 6$$

Question 6.

Concentration of $\text{K}_2\text{Cr}_2\text{O}_7$

$$\begin{aligned} &= \frac{\text{mole}}{\text{volume}} \\ &= \frac{2.5 / 294.185}{0.50} = 0.016\text{M} \end{aligned}$$



n-factor $\text{K}_2\text{Cr}_2\text{O}_7 = 6$

Then normality of $\text{K}_2\text{Cr}_2\text{O}_7 = 0.096$

$$\text{Then } \frac{W}{E} = N \times V$$

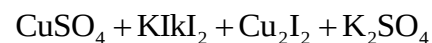
$$\Rightarrow \frac{W}{118.7 / 2} = \frac{0.06 \times 10}{1000}$$

$$W = 0.06 \text{ gm}$$

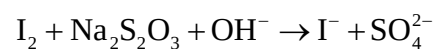
$$\text{Then \% of Sn}^{2+} = \frac{0.06 \times 100}{4} = 15.14\%$$

$$\Rightarrow \frac{x}{17} \Rightarrow 0.89$$

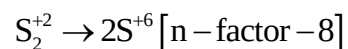
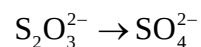
Question 7.



(i) n-factor of $\text{CuSO}_4 = 1$



Equivalent of $\text{I}_2 = \text{equivalent of Na}_2\text{S}_2\text{O}_3$



Then equivalent of I_2

$$= \frac{8 \times 50 \times 1}{1000}$$

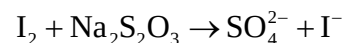
The same equation for $CuSO_4$ which is $\frac{8 \times 50}{1000}$

$$\text{Then } \left(\frac{W}{E} \right)_{CuSO_4} = \frac{8 \times 50}{1000}$$

$$\text{Then } \frac{w}{159.60} = \frac{8 \times 50}{1000}$$

$$W = 63.84$$

$$\text{Then \% of purity} = \frac{83.84 \times 100}{80} = 80\%$$

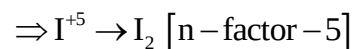
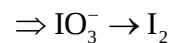


1 M – $Na_2S_2O_3$ then n-factor is 6

$$\text{Hence normality} = 1 \times 8 = 8$$

$$\Rightarrow \text{Hence milliequivalent} = \frac{8 \times V}{1000}$$

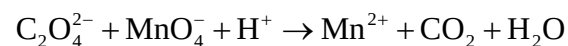
$$\text{Equivalent of } KIO_3 = \frac{W}{E} = 214$$



$$\text{Then } \left[\frac{214}{214} \times 5 \right] = \frac{8 \times V}{1000}$$

$$V = 650$$

(iii) Reaction of $C_2O_4^{2-}$



$$\text{n for } C_2O_4^{2-} = 2$$

$$\text{n for } MnO_4^- = 5$$

$$\text{Milliequivalent of } C_2O_4^{2-} = \text{milliequivalents of } MnO_4^-$$

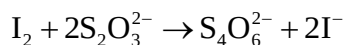
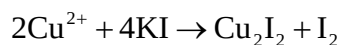
$$\text{m Eq.} = 20 \times \frac{M}{4} \times 5$$

$$\text{m Eq. of } \text{C}_2\text{O}_4^{2-} = 25$$

$$\text{Millimoles of } \text{C}_2\text{O}_4^{2-} = \frac{\text{m Eq.}}{\text{n-factor}} = \frac{25}{2} = 12.5 \text{ millimoles}$$

Cu^{2+} reacts with KI

Reaction of Cu^{2+} :

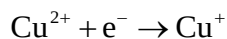


$$\text{m Eq. of } \text{C}_2\text{O}_4^{2-} = 25 \times \frac{M}{10} \times 1 = 2.5 \text{ m Eq.}$$

$$\text{m Eq. of } \text{C}_2\text{O}_4^{2-} = \text{mEq. of } \text{Cu}^{2+}$$

$$\therefore \text{m moles of } \text{Cu}^{2+} = \frac{25}{1} = 2.5 \text{ millimoles}$$

The n-factor of Cu^{2+} is

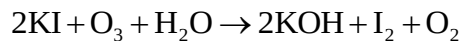


$\therefore$ Difference in the millimoles of $\text{C}_2\text{O}_4^{2-}$ of Cu^{2+}

$$\text{is} = 12.5 - 2.5$$

$$= 10 \text{ millimoles}$$

(iv)



$$\text{M eq. of } \text{I}_2 = \text{Meq. Of } \text{Na}_2\text{S}_2\text{O}_3 = 200 \times 0.1 = 20$$

$$\text{m moles of } \text{I}_2 = \frac{\text{Meq}}{\text{valency}} = \frac{20}{2} = 10$$

Also millimoles of O_3 = millimoles

I_2 (mole ratio = 1 : 1)

Millimoles of O_3 = 10

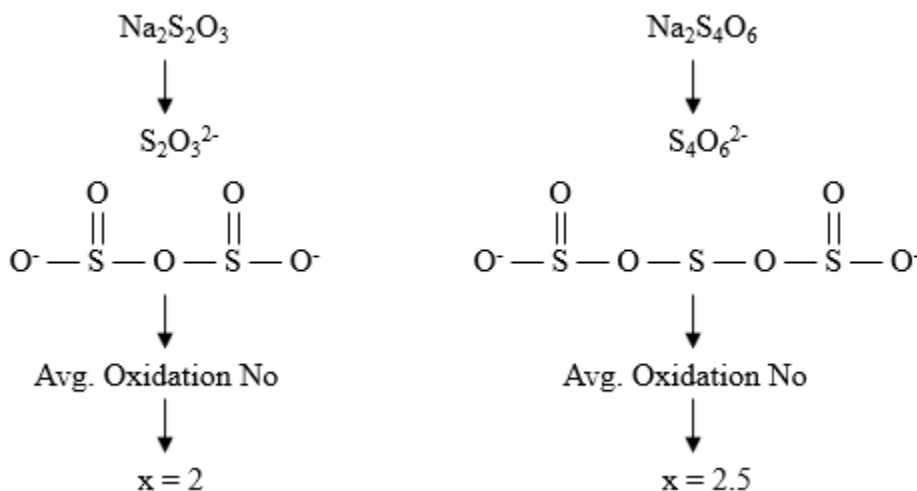
Or moles = 10×10^{-3}

$$D = nRT/V$$

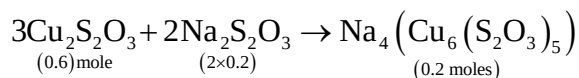
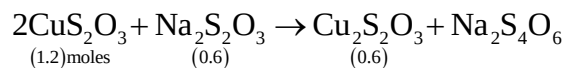
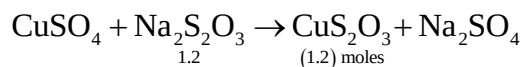
$$PO_3 = \frac{10 \times 10^{-3} \times 0.082 \times 273}{22.4}$$

$$PO_3\% = 10 \times 10^{-3} \times 100 = 1\%$$

Question 8 (i)



Question 8 (ii)

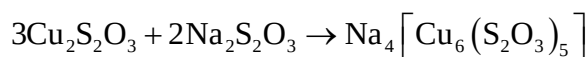
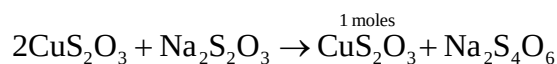
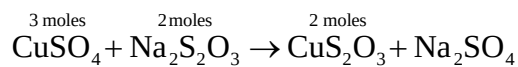


$$\text{Initial moles of } Na_2S_2O_3 = \frac{632}{\text{molar mass of } Na_2S_2O_3}$$

$$\text{Then } Na_2S_2O_3 \text{ reacted} = 1.2 + 0.6 + 0.4 = 2.2$$

$$\text{Not reacted} = 4 - 2.2 = 1.8$$

Question 8 (iii)



As $Na_2S_2O_3$ present in the second reaction is zero hence nothing is present for, $Na_2S_2O_3$. Hence amount formed = 0

